MSG No. 208.44 $P4_232$ [Type I, cubic]

Table 1: Wyckoff site: 2a, site symmetry: 23.

No.	position	mapping
1	[0, 0, 0]	[1,8,9,10,17,18,19,20,21,22,23,24]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,3,4,5,6,7,11,12,13,14,15,16]

Table 2: Wyckoff site: 4b, site symmetry: .32

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,12,14,16,17,18]
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[2,7,9,15,19,24]
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[3,4,10,11,20,21]
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[5,6,8,13,22,23]

Table 3: Wyckoff site: 4c, site symmetry: .32

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[1,12,14,16,17,18]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[2,7,9,15,19,24]
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[3,4,10,11,20,21]
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[5,6,8,13,22,23]

Table 4: Wyckoff site: 6d, site symmetry: 222..

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[1,8,9,10]
2	$\left[\frac{1}{2}, 0, 0\right]$	[2,3,13,14]
3	$\left[0, 0, \frac{1}{2}\right]$	[4,5,15,16]
4	$\left[0, \frac{1}{2}, 0\right]$	[6,7,11,12]
5	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[17,20,22,24]
6	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[18,19,21,23]

Table 5: Wyckoff site: 6e, site symmetry: 2.22

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[1,8,13,14]
2	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[2,3,9,10]
3	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[4,16,18,21]
4	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[5,15,19,23]
5	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[6,11,20,22]
6	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[7,12,17,24]

Table 6: Wyckoff site: `6f`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,8,13,14]`
2	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[2,3,9,10]`
3	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[4,16,18,21]`
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[5,15,19,23]`
5	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[6,11,20,22]`
6	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[7,12,17,24]`

Table 7: Wyckoff site: `8g`, site symmetry: `.3.`

No.	position	mapping
1	$[x, x, x]$	`[1,17,18]`
2	$\left[x+\frac{1}{2}, \frac{1}{2}-x, x+\frac{1}{2}\right]$	`[2,7,15]`
3	$\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-x\right]$	`[3,4,11]`
4	$\left[\frac{1}{2}-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	`[5,6,13]`
5	$[x, -x, -x]$	`[8,22,23]`
6	$[-x, x, -x]$	`[9,19,24]`
7	$[-x, -x, x]$	`[10,20,21]`
8	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	`[12,14,16]`

Table 8: Wyckoff site: `12h`, site symmetry: `2..`

No.	position	mapping
1	$[x, 0, 0]$	`[1,8]`
2	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$	`[4,16]`
4	$\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$	`[5,15]`
5	$\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	`[6,11]`
6	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	`[7,12]`
7	$[-x, 0, 0]$	`[9,10]`
8	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	`[13,14]`
9	$[0, x, 0]$	`[17,24]`
10	$[0, 0, x]$	`[18,21]`
11	$[0, 0, -x]$	`[19,23]`
12	$[0, -x, 0]$	`[20,22]`

Table 9: Wyckoff site: `12i`, site symmetry: `2..`

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	`[1,8]`
2	$\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,3]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}-x\right]$	`[4,16]`
4	$\left[0, \frac{1}{2}, x+\frac{1}{2}\right]$	`[5,15]`
5	$\left[\frac{1}{2}, x+\frac{1}{2}, 0\right]$	`[6,11]`

continued ...

Table 9

No.	position	mapping
6	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	[7,12]
7	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[9,10]
8	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[13,14]
9	$\left[\frac{1}{2},\ x,\ 0\right]$	[17,24]
10	$\left[0,\ \frac{1}{2},\ x\right]$	[18,21]
11	$\left[0,\ \frac{1}{2},\ -x\right]$	[19,23]
12	$\left[\frac{1}{2},\ -x,\ 0\right]$	[20,22]

Table 10: Wyckoff site: 12j, site symmetry: 2..

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	[1,8]
2	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[2,3]
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-x\right]$	[4,16]
4	$\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$	[5,15]
5	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[6,11]
6	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[7,12]
7	$\left[-x,\ \frac{1}{2},\ 0\right]$	[9,10]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[13,14]
9	$\left[0,\ x,\ \frac{1}{2}\right]$	[17,24]
10	$\left[\frac{1}{2},\ 0,\ x\right]$	[18,21]
11	$\left[\frac{1}{2},\ 0,\ -x\right]$	[19,23]
12	$\left[0,\ -x,\ \frac{1}{2}\right]$	[20,22]

Table 11: Wyckoff site: 12k, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ \frac{1}{2}-y\right]$	[1,14]
2	$\left[\frac{3}{4},\ y,\ y+\frac{1}{2}\right]$	[2,9]
3	$\left[\frac{3}{4},\ -y,\ \frac{1}{2}-y\right]$	[3,10]
4	$\left[-y,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[4,21]
5	$\left[y,\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[5,23]
6	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ -y\right]$	[6,22]
7	$\left[y+\frac{1}{2},\ \frac{1}{4},\ -y\right]$	[7,24]
8	$\left[\frac{1}{4},\ -y,\ y+\frac{1}{2}\right]$	[8,13]
9	$\left[y+\frac{1}{2},\ \frac{3}{4},\ y\right]$	[11,20]
10	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ y\right]$	[12,17]
11	$\left[-y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[15,19]
12	$\left[y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[16,18]

Table 12: Wyckoff site: `12l`, site symmetry: `..2`

No.	position	mapping
1	$\left[\frac{1}{4},\, y,\, y+\frac{1}{2}\right]$	`[1,13]`
2	$\left[\frac{3}{4},\, -y,\, y+\frac{1}{2}\right]$	`[2,10]`
3	$\left[\frac{3}{4},\, y,\, \frac{1}{2}-y\right]$	`[3,9]`
4	$\left[y,\, y+\frac{1}{2},\, \frac{1}{4}\right]$	`[4,18]`
5	$\left[-y,\, y+\frac{1}{2},\, \frac{3}{4}\right]$	`[5,19]`
6	$\left[\frac{1}{2}-y,\, \frac{3}{4},\, y\right]$	`[6,20]`
7	$\left[y+\frac{1}{2},\, \frac{1}{4},\, y\right]$	`[7,17]`
8	$\left[\frac{1}{4},\, -y,\, \frac{1}{2}-y\right]$	`[8,14]`
9	$\left[y+\frac{1}{2},\, \frac{3}{4},\, -y\right]$	`[11,22]`
10	$\left[\frac{1}{2}-y,\, \frac{1}{4},\, -y\right]$	`[12,24]`
11	$\left[y,\, \frac{1}{2}-y,\, \frac{3}{4}\right]$	`[15,23]`
12	$\left[-y,\, \frac{1}{2}-y,\, \frac{1}{4}\right]$	`[16,21]`

Table 13: Wyckoff site: `24m`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[x+\frac{1}{2},\, \frac{1}{2}-z,\, y+\frac{1}{2}\right]$	`[2]`
3	$\left[x+\frac{1}{2},\, z+\frac{1}{2},\, \frac{1}{2}-y\right]$	`[3]`
4	$\left[z+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-x\right]$	`[4]`
5	$\left[\frac{1}{2}-z,\, y+\frac{1}{2},\, x+\frac{1}{2}\right]$	`[5]`
6	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[6]`
7	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[7]`
8	$[x,\, -y,\, -z]$	`[8]`
9	$[-x,\, y,\, -z]$	`[9]`
10	$[-x,\, -y,\, z]$	`[10]`
11	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[11]`
12	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	`[12]`
13	$\left[\frac{1}{2}-x,\, z+\frac{1}{2},\, y+\frac{1}{2}\right]$	`[13]`
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-z,\, \frac{1}{2}-y\right]$	`[14]`
15	$\left[z+\frac{1}{2},\, \frac{1}{2}-y,\, x+\frac{1}{2}\right]$	`[15]`
16	$\left[\frac{1}{2}-z,\, \frac{1}{2}-y,\, \frac{1}{2}-x\right]$	`[16]`
17	$[z,\, x,\, y]$	`[17]`
18	$[y,\, z,\, x]$	`[18]`
19	$[-y,\, z,\, -x]$	`[19]`
20	$[-z,\, -x,\, y]$	`[20]`
21	$[-y,\, -z,\, x]$	`[21]`
22	$[z,\, -x,\, -y]$	`[22]`
23	$[y,\, -z,\, -x]$	`[23]`
24	$[-z,\, x,\, -y]$	`[24]`